



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

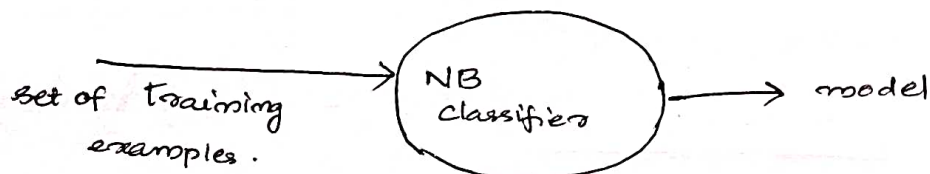
 Website: www.ktunotes.in

Module-2

Naive Bayes classifier.

decision Tree

- * Naive bayes classifier:- it is a probability based classifier.
It is based on bayes theorem



model:- model is used to predict the target value of new instance in terms of probability.

Training phase:- Various $P(V_j)$ & $P(A_i|V_j)$ terms are estimated using the training data.

$V_j \rightarrow$ target Concept Value
eg:- {yes, no}

$A_i \rightarrow$ Attribute Value

outlook = {sunny, Overcast, Rain}

Temp = {Hot, Mild, Cool}

Humidity = {High, Normal}

wind = {Weak, Strong}

Play? = {No, Yes}

(probabilities)

The set of these estimates corresponds to learned hypothesis

Testing phase:- These learned hypothesis then used to classify each new instance by applying

$$\underset{V_j \in V}{\text{Arg Max}} \left\{ P(V_j) \cdot \prod_{i=1}^n P(A_i|V_j) \right\}$$

given,

the below data set apply Naive bayes classification

Here,

The target attribute is play tennis which can have values Yes or no. for different days.

Each day described by the attributes

<outlook, Temp, Humidity, Wind>

Here,

The attribute values of outlook are. Sunny, Rainy, Overcast

The attribute values of temp. are {Hot, Mild, Cool}

The attribute values of humidity are {High, Normal}

The attribute values of wind are weak, strong.

Learning phase:

$P(\text{play tennis}) =$

$$P(\text{play tennis} = \text{yes}) = 9/14$$

$$P(\text{play tennis} = \text{no}) = 5/14$$

$$P(B|A) = \frac{P(B/A) * P(A)}{P(B)}$$

~~$P(A)$~~ $P(\text{Outlook} = \text{Sunny} / \text{yes}) = ?$ ~~$2/14$~~

~~$P(A)$~~

$$= \frac{P(\text{Yes} / \text{outlook} = \text{Sunny}) * P(\text{outlook} = \text{Sunny})}{P(\text{Yes})}$$

$$= \frac{2/5 * 5/14}{9/14}$$

$$= \underline{\underline{2/9}}$$

$$P(\text{overcast} / \text{yes}) = \frac{P(\text{Yes} / \text{overcast}) * P(\text{overcast})}{P(\text{Yes})}$$

$$= \frac{4/4 * 4/14}{9/14}$$

$$= \underline{\underline{4/9}}$$

$$P(\text{rain} / \text{yes}) = \frac{3}{9} = \frac{1}{3} //$$

$$P(\text{sunny} / \text{no}) = \frac{3}{5}$$

Tuesday
11/10/2022

$$P(\text{overcast} / \text{no}) = \frac{0}{5}$$

$$P(\text{Rain} / \text{no}) = \frac{2}{5}$$

Temp.

$$* P(\text{Temp} = \text{hot} / \text{yes}) = \frac{2}{9} /$$

$$P(\text{mild} / \text{yes}) = \frac{4}{9} / 0.55$$

$$P(\text{cool} / \text{yes}) = \frac{3}{9} = \frac{1}{3} /$$

$$P(\text{hot} / \text{no}) = \frac{2}{5} /$$

$$P(\text{mild} / \text{no}) = \frac{2}{5} /$$

$$P(\text{cool} / \text{no}) = \frac{1}{5}$$

Humidity

$$P(\text{humidity} = \text{High} / \text{yes}) =$$

$$P(\text{Normal} / \text{yes}) = ? \quad 6/9$$

$$P(\text{High} / \text{No}) = ? \quad 3/5$$

$$P(\text{Normal} / \text{No}) = ? \quad 4/5$$

$$\frac{P(\text{yes} / \text{hot}) * P(\text{hot})}{P(\text{yes})} = \frac{\frac{2}{9} * \frac{4}{9}}{\frac{9}{14}}$$

$$\frac{P(\text{yes} / \text{mild}) * P(\text{mild})}{P(\text{yes})} = \frac{* \frac{6}{14}}{\frac{9}{14}}$$

Day	Outlook	Temp.	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Wind

$$P(\text{Wind} = \text{weak} / \text{Yes}) = 8/9$$

$$P(\text{Wind} = \text{strong} / \text{Yes}) = 3/9$$

$$P(\text{Wind} = \text{weak} / \text{No}) = 2/5$$

$$P(\text{Wind} = \text{strong} / \text{No}) = 3/5$$

TESTING PHASE

Let $\langle \text{Sunny, Cool, high, strong} \rangle$? $\rightarrow$ target
Values $\rightarrow$ test data

Let,

Sunny, Cool, high, Strong be test data. then the NB classifier (calculate) the target class as follows

$$\begin{aligned} \textcircled{1} & P(\text{Yes}) * P(\text{Sunny} / \text{Yes}) * P(\text{Cool} / \text{Yes}) * P(\text{high} / \text{Yes}) * P(\text{Strong} / \text{Yes}) \\ &= \frac{9}{14} * \frac{2}{9} * \frac{1}{3} * \frac{3}{9} * \frac{3}{9} = \frac{1}{189} \approx 0.0053 \end{aligned}$$

$$\begin{aligned} \textcircled{2} & P(\text{No}) * P(\text{Sunny} / \text{No}) * P(\text{Cool} / \text{No}) * P(\text{high} / \text{No}) * P(\text{Strong} / \text{No}) \\ &= \frac{5}{14} * \frac{3}{5} * \frac{1}{5} * \frac{3}{5} * \frac{3}{5} \\ &= \frac{27}{1750} \approx 0.0154 \end{aligned}$$

Here,

$$\text{Arg} \max_{V_j \in Y} \left(P(V_j) \cdot \prod_{i=1}^n P(a_i / V_j) \right)$$

Here,

probability of play tennis = No $\Rightarrow 0.0206$

is greater than probability of play tennis = Yes.

So, the classifier predict play tennis = No

Q/ let,

<overcast, Mild, Normal, Weak, ?>

$$\textcircled{*} P(\text{Yes}) * P(\text{overcast}/\text{Yes}) * P(\text{mild}/\text{Yes}) * P(\text{Normal}/\text{Yes}) * P(\text{weak}/\text{Yes})$$
$$\frac{9}{14} * \frac{4}{9} * \frac{4}{9} * \frac{6}{9} * \frac{6}{9} = \frac{32}{567} = 0.056$$

$$\textcircled{*} P(\text{no}) * P(\text{overcast}/\text{no}) * P(\text{mild}/\text{no}) * P(\text{Normal}/\text{No}) * P(\text{weak}/\text{No})$$
$$= \frac{5}{14} * 0 * \frac{2}{5} * \frac{1}{5} * \frac{2}{5}$$
$$= 0 //$$

using data sets:

$P(\text{Yes})$	$P(\text{mild}/\text{Yes})$	$P(\text{weak}/\text{Yes})$
$P(\text{no})$	$P(\text{mild}/\text{No})$	$P(\text{Weak}/\text{No})$
$P(\text{overcast}/\text{Yes})$	$P(\text{Normal}/\text{Yes})$	
$P(\text{overcast}/\text{No})$	$P(\text{Normal}/\text{No}) = \frac{1/4 + 7/14}{9/14} = 1/5$	

Decision Tree

+ It is a non-linear model that performs both regression and classification.

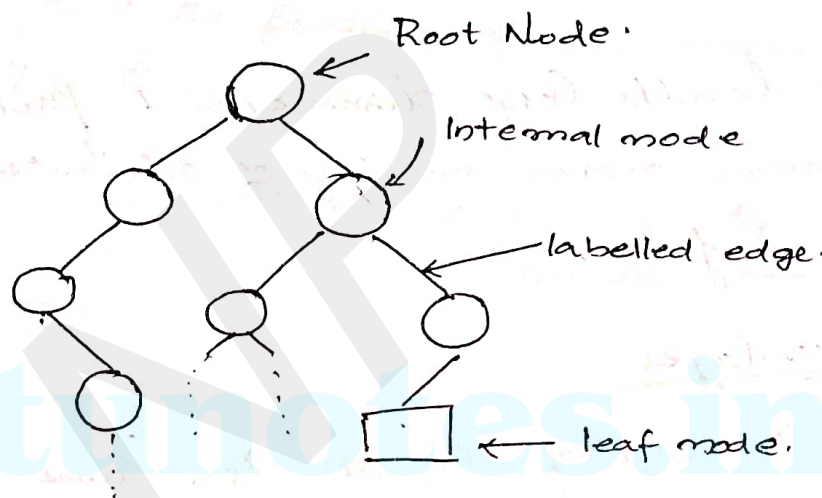
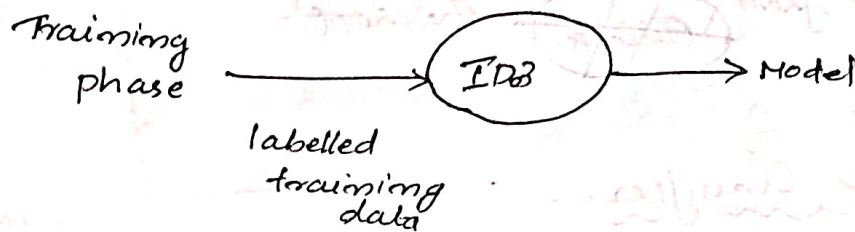
- It uses supervised method

Working

learning/training phase.

In the training phase a set of training examples is given to the decision tree learning algorithm. [ID3]

Then the algorithm creates a decision tree model. This model can be used for regression or classification.

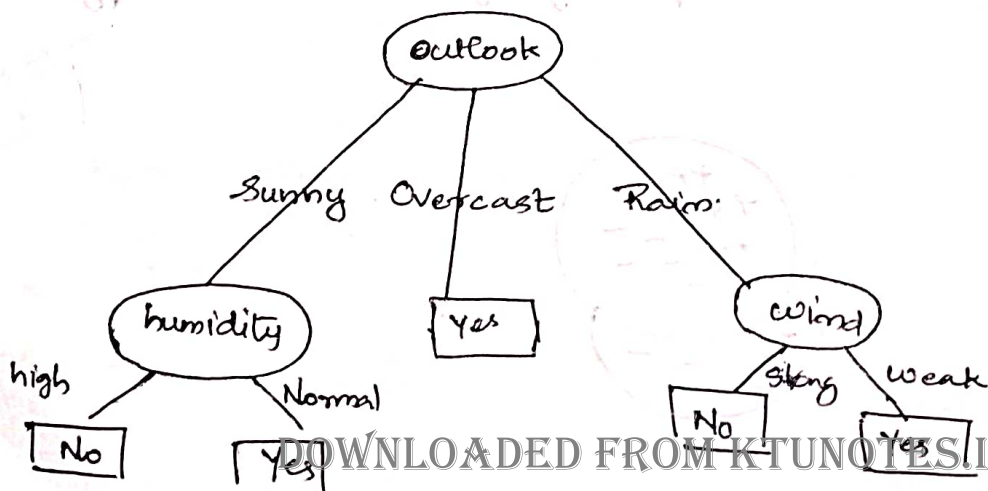


- * Each Internal node is labelled with an attribute [best attribute]
- * That best splits the training data.

In order to find such an attribute entropy and information gain is used.

- * Leaf node contains target labels.
- * Edges are labelled with attribute values

For example:



wed
12/10/2022

Testing phase

decision tree At the time of instance new instance is given to the model then the model finds the target value of that instance by traversing the tree starting from ~~Outlook~~ the root.

Advantages

① Non-linear Classifier.

It can be used for complex classification and regression class.

② It can handle large dimensional problems.
dimension means number of attributes.

③ testing is faster.

Disadvantages

① * As the size of the tree increases time complexity also increases.

② * Overfitting and Underfitting problems.

* Entropy and Information gain

* ID3 Algorithm uses the information gain measure to find the best attribute.

Entropy:-

It measures the impurity of a collection of examples



gain,

a collection S containing +ve and -ve examples of some target concept. Then the entropy of the samples

$$\text{Entropy}(S) = -P_{+} \log_2 P_{+} - P_{-} \log_2 P_{-}$$

where,

$P_{+} \rightarrow P_{+}$ is the proportion of +ve examples in S
 $P_{-} \rightarrow$ It is the proportion of -ve examples in S

eg:- Suppose S is a collection of 14 examples including 9 positive and 5 negative examples. Find the entropy of S .

$$\text{Entropy}(S) = -P_{+} \log_2 P_{+} - P_{-} \log_2 P_{-}$$

$$E(S) = -\frac{9}{14} \log_2 \frac{9}{14}$$

$$- \frac{5}{14} \log_2 \frac{5}{14}$$

$$= \left(-\frac{9}{14} \log_2 \frac{9}{14} \right) - \left(\frac{5}{14} \log_2 \frac{5}{14} \right)$$

$$= \underline{0.94028}$$

$$P_{+} \rightarrow \frac{9}{14}$$

$$P_{-} \rightarrow \frac{5}{14}$$

Information Gain

Gain(S, A) if an attribute A relative to a collection of samples S is defined as

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \times \text{Entropy}(S_v)$$

Values(A) $\Rightarrow$

Set of all possible values for attribute A

$S_v \rightarrow$ it is the subset of S for which attribute A has value V (S $\rightarrow$ total collection of examples.)

$$= \underbrace{\text{Entropy}(S)}_{\text{1.}} - \underbrace{\sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \times \text{Entropy}(S_v)}_{\text{2.}}$$

this term gives the entropy of original collection S. It gives the expected value of entropy after S is partitioned using A

Q) Find the information gain of wind attribute

$$\text{Gain}(S, \text{wind}) = E(S) - \sum_{v \in \{\text{weak}, \text{strong}\}} \frac{|S_v|}{|S|} \times E(S_v)$$

$$= E(S) - \left[\frac{|S_{\text{weak}}|}{|S|} \times E(S_{\text{weak}}) + \right.$$

$$\left. \frac{|S_{\text{strong}}|}{|S|} \times E(S_{\text{strong}}) \right]$$

$$E(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$|S| = 14$$

$$|S_{\text{weak}}| = 8$$

$$|S_{\text{strong}}| = 6$$

1) Find out Target class of below Test data.

(1) < Rain, Mild, High, Strong, ? >

(2) Find Entropy (S_{weak})

$$\begin{aligned} E(s) &= -P_{\oplus} \log_2 P_{\oplus} - P_{\ominus} \log_2 P_{\ominus} \\ &= 1 \end{aligned}$$

$$P(Yes) * P(Rain/Yes) * P(Mild/Yes) * P(High/Yes) * P(Strong/Yes)$$

$$= \frac{1}{3} * \frac{4}{9} * \frac{6}{9} * \frac{3}{9} = \frac{8}{243} = 0.0329$$

$$P(no) * P(Rain/no) * P(Mild/no) * P(High/no) * P(Strong/no)$$

$$= \frac{2}{5} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{36}{625} = 0.0576$$

Thursday
12/6/2022

$$\text{Gain}(S, \text{Wind}) = \text{Entropy}(S) - \left[\frac{|S_{weak}|}{|S|} * E(S_{weak}) + \frac{|S_{strong}|}{|S|} * E(S) \right]$$

$$|S| = 14$$

$$|S_{weak}| = 8$$

$$|S_{strong}| = 6$$

general equation to find entropy.

$$\text{Entropy}(S) = -P_{\oplus} \log_2 P_{\oplus} - P_{\ominus} \log_2 P_{\ominus}$$

$$\text{Entropy}(S) = 0.12073$$

$$= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$= 0.12073$$

$$E(S_{\text{weak}}) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$$

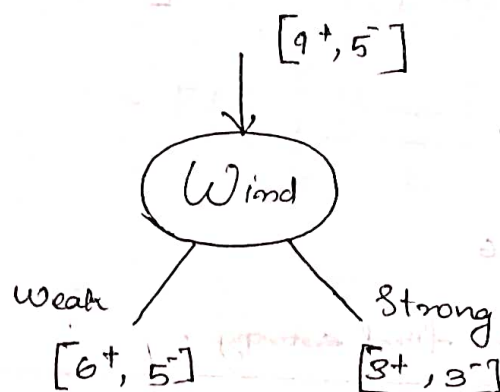
$$= \underline{\underline{0.81124}}$$

$$E(S_{\text{strong}}) = \frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6}$$

$$= \underline{\underline{1}}$$

$$\begin{aligned} \text{Gain}(S, \text{wind}) &= \text{Entropy}(S) - \left[\frac{|S_{\text{weak}}|}{|S|} * E(S_{\text{weak}}) + \frac{|S_{\text{strong}}|}{|S|} * E(S) \right] \\ &= 0.940 - \left[\frac{8}{14} * 0.811 + \frac{6}{14} * 1 \right] \\ &= \underline{\underline{0.048}} \end{aligned}$$

Splitting of training data. using Wind attribute



Q) Using the below data set find the root attribute of the decision tree.

Logic findout the gain of each attribute then select the attribute that having highest gain as the root attribute.

$$\text{Gain}(S_{\text{humidity}}) = \text{Entropy}(S) - \left[\frac{|S_{\text{high}}|}{|S|} * E(S_{\text{high}}) + \frac{|S_{\text{normal}}|}{|S|} * E(S_{\text{normal}}) \right]$$

$$\text{Gain}(S_{\text{humidity}}) = E(S) - \sum_{v \in \{\text{high, Normal}\}} \frac{|S_v|}{|S|} * E(S_v)$$

$$= E(S) - \left[\frac{|S_{\text{high}}|}{|S|} * E(S_{\text{high}}) + \frac{|S_{\text{normal}}|}{|S|} * E(S_{\text{normal}}) \right]$$

$$\begin{aligned} |S| &= 14 \\ |S_{\text{high}}| &= 7 \\ |S_{\text{normal}}| &= 7 \\ E(S) &= 0.940 \end{aligned}$$

$$= 0.940 - \left[\frac{7}{14} * 0.985 + \frac{7}{14} * 0.591 \right]$$

$$= 0.152$$

$$\underline{\underline{0.1518}}$$

$$\underline{\underline{0.1518}}$$

$$E(S_{\text{high}}) = \left(-\frac{3}{7} \log_2 \frac{3}{7} \right) - \left(\frac{4}{7} \log_2 \frac{4}{7} \right)$$

$$= 0.9852$$

$$E(S_{\text{normal}}) = \left(-\frac{6}{7} \log_2 \frac{6}{7} \right) - \left(\frac{1}{7} \log_2 \frac{1}{7} \right)$$

$$= 0.5916$$

$$\text{Gain}(S, \text{outlook}) = E(S) - \sum_{v \in \{\text{Sunny, Overcast, Rain}\}} \frac{|S_v|}{|S|} * E(S_v)$$

$$\text{Gain}(S, \text{outlook}) = E(S) - \left[\frac{|S_{\text{sunny}}|}{|S|} * E(S_{\text{sunny}}) + \frac{|S_{\text{overcast}}|}{|S|} * E(S_{\text{overcast}}) + \frac{|S_{\text{rain}}|}{|S|} * E(S_{\text{rain}}) \right]$$

~~2.15~~

$$E(\text{Sunny}) = \left(-\frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{3}{5} \log_2 \frac{3}{5} \right)$$

$$= \underline{0.9709}$$

$$E(\text{overcast}) = \left(-\frac{4}{4} \log_2 \frac{4}{4} \right) - \left(\frac{0}{4} \log_2 \frac{0}{4} \right)$$

$$= 0$$

$$E(\text{Rain}) = \left(-\frac{3}{5} \log_2 \frac{3}{5} \right) - \left(\frac{2}{5} \log_2 \frac{2}{5} \right)$$

$$= \underline{0.9709}$$

$$|S| = 4$$

$$|\text{Sunny}| = 5$$

$$|\text{overcast}| = 4$$

$$|\text{Rainy}| = 5$$

$$E(S) = 0.9402$$

$$0.9402 = \left[\frac{5}{4} \times 0.9709 + \frac{4}{4} \times 0 + \frac{5}{4} \times 0.9709 \right]$$

=

Gain (S, Temp) $\Rightarrow$

$$E(S) = \frac{|S_{\text{Hot}}|}{|S|} \times E(S_{\text{Hot}}) + \frac{|S_{\text{Mild}}|}{|S|} \times E(S_{\text{Mild}}) + \frac{|S_{\text{Cool}}|}{|S|} \times E(S_{\text{Cool}})$$

$$\Rightarrow E(S) = 0.9402 \quad |S| = 14$$

$$|S_{\text{Hot}}| = 4$$

$$|S_{\text{Cool}}| = 4$$

$$|S_{\text{Mild}}| = 6$$

$$E(S_{hot}) = \frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4}$$

$$= 1 //$$

$$E(S_{mild}) = \frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6}$$

$$= 0.91 //$$

$$E(S_{cool}) = \frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$= 0.81 //$$

$$E(S)$$

$$0.97 - \left(\frac{1}{14} \times 1 + \frac{6}{14} \times 0.91 + \frac{4}{14} \times 0.81 \right)$$

$$= 0.032 //$$

$$0.032 //$$

Friday
14/6/2022

$$\text{Gain}(S, \text{outlook}) = 0.24 \checkmark$$

$$\text{Gain}(S, \text{wind}) = 0.48 \checkmark$$

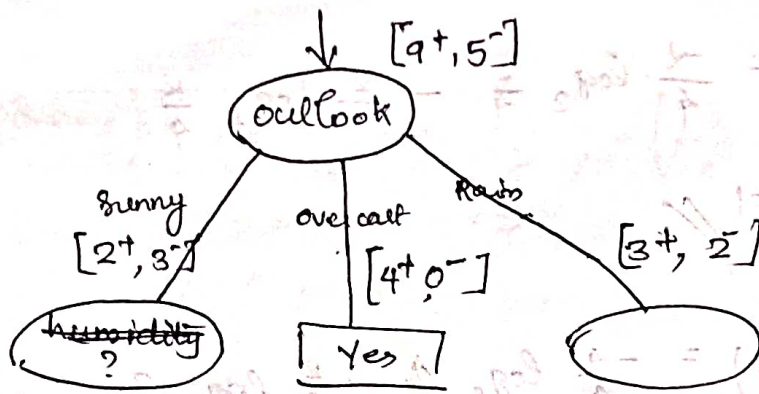
$$\text{Gain}(S, \text{temp}) = 0.082 \checkmark$$

$$\text{Gain}(S, \text{humidity}) = 0.15 \checkmark$$

hence,

the outlook attribute has the max gain. This is the root attribute of the decision tree

Splitting of the training data



Imp

Basic decision tree learning algorithm [ID3]

ID3 is the most popular method of generating decision tree from the training data. The tree is constructed from top to bottom.

1. $A \leftarrow$ find the best decision tree attribute, based on the training data D .
2. Create a node labelled A .
3. For each attribute value of A , a child node is created.
4. Sort the training examples, to these child nodes, based on the attribute value of branch.
5. If all the training examples are perfectly classified then ~~stop~~ stop and label the node as leaf, with the target value of target attribute.
6. Else, Goto Step 1 and Repeat.

The central choice of ID3 algorithm is finding the best attribute node.

A statistical property called Information gain is used for it.

general equations of Information Gain.

$$\text{Gain}(S, A) = E(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} * E(S_v)$$

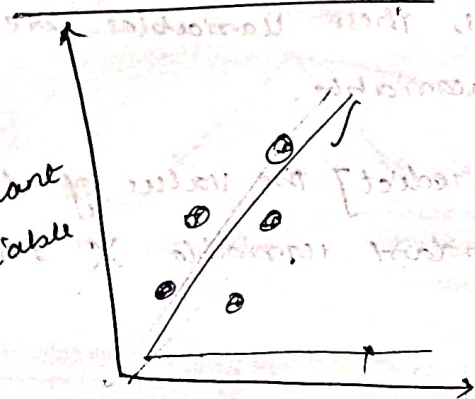
Entropy

$$\text{Entropy}(S) = -P_0 \log_2 P_0 - P_1 \log_2 P_1$$

Regression

linear model

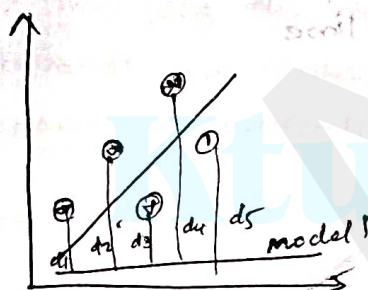
Price
dependant
variable



Milage

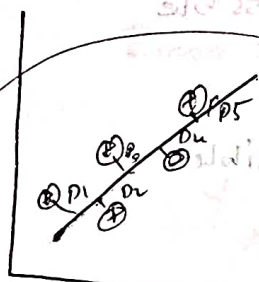
independent
variable.

* Linear regression with one variable



model 1

$$\sum_{i=1}^n d_i^2 = d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$



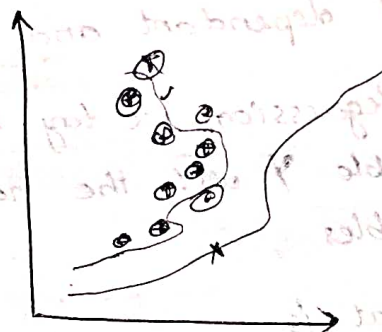
model 2

$$\sum_{i=1}^n d_i^2 = d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

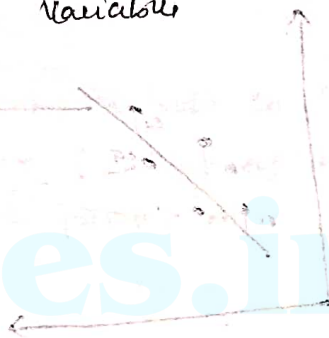
minimum distance.

best regression model.

non-linear model



* Linear regression with multiple variables



Tuesday
18/10/2022

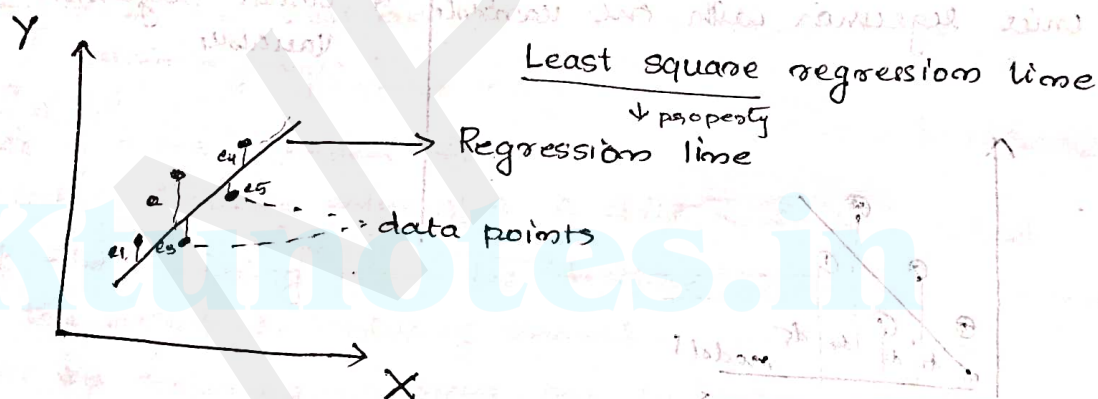
Regression

* It is a statistical method used to analyze the and understand relationship b/w two or more variables, these variables are called dependant and independant variable.

* In Regression we try to regress [predict] the value of dependant variable Y with the help of independant variable X or variables

That is,

We try to understand how the value of Y changes with respect to change in X



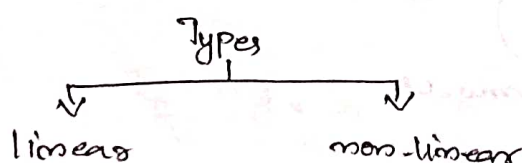
The least square regression line is a unique line such that,
* Sum of the squared vertical distance (Y) between the actual data points and line is smallest as possible

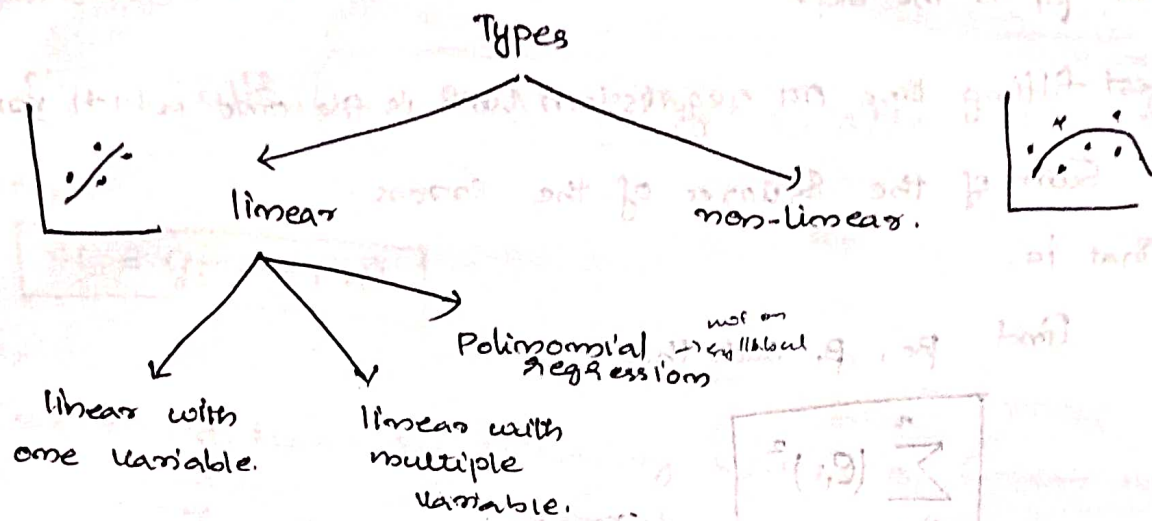
$$E = y = e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2$$

Should be minimum as possible.

A ML algorithm automatically finds such a regression line after proper training

Types of Regression





Linear Regression with one variable

independent

* It is characterized by one independent variable
eg:-

Predicting the price of a house based on the size of the plot.

In this case the regression problem is use to find out the relationship b/w dependent variable (price of house) and independent variable (size of the plot)

Let,

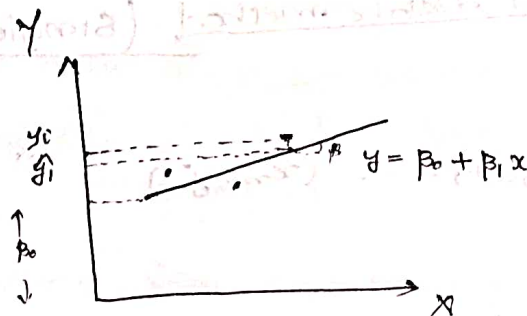
data points: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Model: $y = \beta_0 + \beta_1 x$

Model prediction: $\hat{y}_i = \beta_0 + \beta_1 x_i$

Error: $e_i = y_i - \hat{y}_i$

Actual Predicted



where,

β_0 is the y intercept

and β_1 is the slope

A best fitting line or regression line is the one which minimizes
~~Sum~~ Sum of the squares of the errors.

That is,

Find β_0, β_1 such that

$$\sum_{i=1}^n (e_i)^2$$

errors term

errors term should be minimum as possible.

i.e.,

minimize

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

So a machine learning algorithm tries to learn

β_0, β_1 such that it will minimize

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

LMS method is used for this.

Machine learning approach for

Linear Regression with Matrix method (Single Variable)

Let,

$$(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$$

be the data points.

Let,

$$y = f(x) = b + mx$$

is the regression line, which will minimize the sum of

the squared error.

$$\underbrace{(y_1 - f(x_1))^2}_{e_1} + \underbrace{(y_2 - f(x_2))^2}_{e_2} + \underbrace{(y_3 - f(x_3))^2}_{e_3} + \dots + \underbrace{(y_n - f(x_n))^2}_{e_n}$$

where,

$$e_i = y_i - f(x_i)$$

So,

we would have the following system of linear equations:

$$\begin{aligned} y_1 &= f(x_1) + e_1 = b + mx_1 + e_1 \\ y_2 &= f(x_2) + e_2 = b + mx_2 + e_2 \\ &\vdots \\ y_n &= f(x_n) + e_n = b + mx_n + e_n \end{aligned}$$

ie,

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = X \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = XA$$

$$E = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

ie,

$$Y = AX + E$$

now solve it for

$$A = (X^T X)^{-1} X^T Y$$

Sum of squared error

$$SSE = E^T E$$

① determine least square regression line using matrix method

data

Price(x)	Demand(y)
49	124
69	95
89	71
99	45
109	18

Here,

The dependant variable is demand and independant variable is Price(x)
we need to find the relationship b/w price and demand

$$Y = \begin{bmatrix} 124 \\ 95 \\ 71 \\ 45 \\ 18 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 49 \\ 1 & 69 \\ 1 & 89 \\ 1 & 99 \\ 1 & 109 \end{bmatrix} \quad A = \begin{bmatrix} b \\ m \end{bmatrix}_{2 \times 1}$$

$$A = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 49 & 69 & 89 & 99 & 109 \end{bmatrix} \quad \begin{bmatrix} 1 & 49 \\ 1 & 69 \\ 1 & 89 \\ 1 & 99 \\ 1 & 109 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 415 \\ 415 & 36765 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{11600} \begin{bmatrix} 36765 & -415 \\ -415 & 5 \end{bmatrix}$$

$$A = (X^T X)^{-1} X^T Y$$

$$= \frac{1}{11600} \begin{bmatrix} 36765 & -415 \\ -415 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 49 & 69 & 89 & 99 & 109 \end{bmatrix} \begin{bmatrix} 124 \\ 95 \\ 71 \\ 45 \\ 18 \end{bmatrix}$$

$$= \frac{1}{11600} \begin{bmatrix} 36765 & -415 \\ -415 & 5 \end{bmatrix} \cdot \begin{bmatrix} 353 \\ 25365 \end{bmatrix}$$

$$= \begin{bmatrix} 211 \\ -1.7 \end{bmatrix}$$

$$\approx \begin{bmatrix} b \\ m \end{bmatrix}$$

$$\hat{y} = 211 - 1.7x$$

(ii) Find Sum of squared ^{error} element

$$SSE = E^T E$$

X	Y	$f(x_i) = 211 - 1.7x_i$	$e_i = y_i - f(x_i)$
49	124	127.7	-3.7
69	95	98.7	1.3
89	71	59.7	11.3
99	45	42.7	2.3
109	18	25.7	-7.7

$$E = \begin{bmatrix} -3.7 \\ 1.3 \\ 11.3 \\ 2.3 \\ -7.7 \end{bmatrix}$$

$$SSE = E^T E$$

$$= \begin{bmatrix} -3.7 & 1.3 & 11.3 & 2.3 & -7.7 \end{bmatrix}$$

$$= 207.65$$

Linear regression with multiple variables [multi variated regression]

A regression model that estimate the relationship b/w multiple independent variable and one dependent variable.

Let,

$$y = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_n x_{in} + \epsilon_j$$

is the regression model.

where,

y is the dependent variable

$\beta_0 \rightarrow y$ Intercept

$\beta_1 \rightarrow$ slope of the x_1 variable.

$\beta_2 \rightarrow$ slope of x_2 variable.

$\vdots$

$\beta_n \rightarrow$ slope of x_n variable [Regression]

$\epsilon_j \rightarrow$ model errors.

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \dots + \beta_n x_{1n} + \epsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \dots + \beta_n x_{2n} + \epsilon_2$$

$\vdots$

$$y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \dots + \beta_n x_{nn} + \epsilon_n$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1n} \\ 1 & x_{21} & \dots & x_{2n} \\ 1 & x_{31} & \dots & x_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nn} \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

Ques
19/10/2022

Find a multi linear regression model to the following Data set

$$\beta = (X^T X)^{-1} X^T Y$$

dataset

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

y	x ₁	x ₂
3	2	1
2	3	5
4	6	3
6	7	6
8	8	7

$$3 = 1\beta_0 + \beta_1 2 + \beta_2 1$$

$$2 = 1\beta_0 + \beta_1 3 + \beta_2 5$$

$$4 = 1\beta_0 + \beta_1 6 + \beta_2 3$$

$$6 = 1\beta_0 + \beta_1 7 + \beta_2 6$$

$$8 = 1\beta_0 + \beta_1 8 + \beta_2 7$$

$$Y = \begin{bmatrix} 3 \\ 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 5 \\ 1 & 6 & 3 \\ 1 & 7 & 6 \\ 1 & 8 & 7 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Intercept

Slope

$$\beta = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 5 & 25 & 22 \\ 25 & 151 & 130 \\ 22 & 130 & 120 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 1.201 & -0.138 & -0.071 \\ -0.138 & 0.114 & -0.098 \\ -0.071 & -0.098 & 0.128 \end{bmatrix}$$

$$(X^T X)^{-1} X^T Y = \begin{bmatrix} 0.50 \\ 1 \\ -0.25 \end{bmatrix}$$

$$\beta_0 = 0.50$$

$$\beta_1 = 1$$

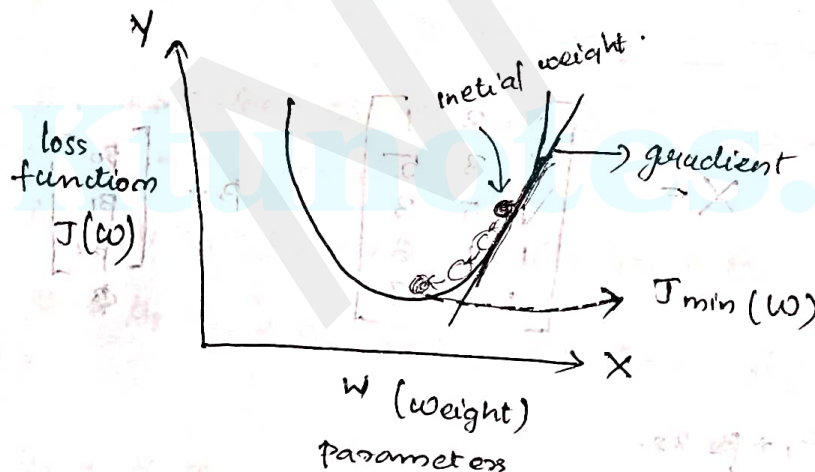
$$\beta_2 = -0.25$$

$$y = 0.50 + x_1 - 0.25 x_2$$



Gradient descent Approach

- * Gradient descent is an optimization algorithm used for minimizing the loss or cost function in various machine learning algorithms.



- * Gradient descent algorithm starts from the top of the mountain (hill) [solution space] then climb down the mountain by updating parameters to learn (weights) and tries to reach at the valley [where cost function is minimum]

Significance of gradient

At each step while climbing down the hill, the gradient (slope) helps to determine the direction of step to be taken to reach at valley.

Significance of learning rate η

It determines the length of the step. That is how fast it should reach at valley

Gradient descent Algorithm

1. Initialize the weights parameters and define loss function. $J(w)$
2. At each step:

* $\rightarrow$ Calculate gradient

$$\Delta W_j = -\eta \times \frac{\partial (J(w))}{\partial W_j}$$

Annotations:
 - ΔW_j is labeled "change in weight"
 - η is labeled "learning rate"
 - $\frac{\partial (J(w))}{\partial W_j}$ is labeled "gradient"

* $\rightarrow W_j = W_j + \Delta W_j$

8. Step 2 is repeated until global cost minimum $J_{\min}(w)$ reached.

Today
21/10/2022

Gradient descent approach for solving linear regression

In the case of a linear regression, the loss function is sum of squared error.

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Annotation: $\rightarrow$ SSE

we need to learn the weight parameters β_0, β_1 in such a way that for suitable values of β_0, β_1 the loss function should be minimum as possible.

Gradient descent approach is used for this task.

1. define the cost function

$$J(\theta) = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Annotation: $\downarrow$
model parameters

$$\beta_0, \beta_1$$

then initialize β_0, β_1 with some value. For

2. For each training example

calculate change in weight

ie, $\Delta \theta_j = -\eta \frac{\partial J(\theta)}{\partial \theta_j}$

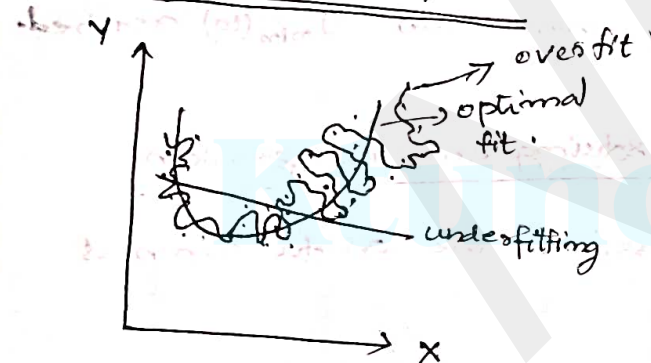
update weights : $\theta_j = \theta_j + \Delta \theta_j$ delta rule.

3. Repeat step 2 until $J(\theta)$ reaches global cost minimum

Overfitting in Regression

Overfitting is a situation in which the model learns too much using the training data. but if the data size is small or data having noise then the learned model performs very well on the testing using training data. but performance will be low for actual test data

Regression example

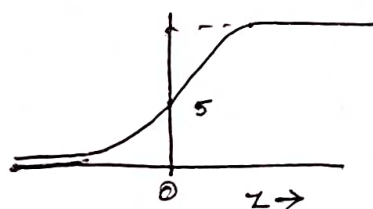


Underfit — Suppose data is taken from a second degree polynomial then try to fit a 1st degree polynomial (Simple model) results in underfitting

over fit :- trying to fit too complex models. [fourth degree polynomial etc]

Logistic Regression

- * It is a ^{binary} classification technique that uses the concept of regression.
- * It uses logistic function [Sigmoid function] for classification.
- ~~* It uses~~ Sigmoid function



$$f(z) = \frac{1}{1 + e^{-z}}$$

$$z \rightarrow \infty$$

$$f(z) \rightarrow 1$$

$$z \rightarrow -\infty$$

$$f(z) \rightarrow 0$$